

QSARena: top-10 on 35 of 37 ADMET benchmarks

One leakage-controlled pipeline, 44 datasets, 28 models - and no architecture dominates.

BENCHMARK



SMILES
to property

44 datasets

5 suites

28 models

TDC | MoleculeNet | Polaris
PODUAM | ChemML

PUBLISHED-REFERENCE TOP-10 PLACEMENT

Best model, chosen on the test set



22/22 on official TDC splits - median rank 3

Model chosen by cross-validation only



median rank 8 - the honest estimate for new data

-10
placements lost
to honest model
selection

NO SINGLE WINNER

Ensembles  **16**

3D pretrained  **11**

Conventional ML  **8**

Top family takes only 36% of datasets;
the best model is dataset-dependent.

RUNS ON A LAPTOP GPU

-0.2% median change vs an A100,
37 identically split datasets

Start with descriptor-rich gradient boosting plus averaging; add 3D pretrained models where chemistry and compute justify it.

github.com/ScottCoffin/QSARena